

HMRP: Hierarchical Concept Mapping for Multidisciplinary Research Paper Similarity Analysis

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Abstract—The volume of scientific literature makes manual comparison of papers slow, and keyword search alone misses conceptual relationships between documents. This paper presents HMRP, a locally executed pipeline for research-paper analysis. HMRP converts PDFs to structured text, extracts keywords with TF-IDF, KeyBERT and YAKE, extracts concepts with spaCy named-entity recognition, technical-term patterns and a fine-tuned SciBERT acronym tagger, and organises the concepts into a domain-taxonomy hierarchy. Papers are compared with TF-IDF cosine similarity (lexical) and sentence-embedding cosine similarity (semantic), and the results are explored through a Streamlit dashboard with sunburst, treemap, icicle, mind-map, knowledge-graph and heatmap views, fuzzy search and multi-format export. Each extracted concept is linked to a verbatim sentence and page, and an eight-rule validator blocks cross-paper contamination. The dataset design spans 15 research domains with 50 papers per domain (750 papers), which the authors report as downloaded. The automated evaluation reported here was run on an earlier validated 50-paper arXiv corpus (1,123 pages, 35/7/8 paper-level split). Against a unigram-frequency baseline HMRP produced 60.0 instead of 15.0 concepts per paper and 100% evidence traceability, and mean pairwise semantic similarity on the 8-paper test split was 0.2424. The generated hierarchy is a computational structure and has not been validated by domain experts.

Index Terms—Hierarchical concept mapping, research paper similarity, concept extraction, keyword extraction, semantic similarity, sentence transformers, information retrieval, knowledge graph, multidisciplinary research.

I. INTRODUCTION

Scientific output grows every year, and a researcher who surveys a topic must compare dozens of papers for shared methods, datasets and ideas. Bibliographic search tools typically match query terms against titles, abstracts and keywords. Lexical weighting schemes such as TF-IDF [1], [2] work well for this purpose but reward shared vocabulary rather than shared meaning: two papers that describe the same idea in different words score poorly, while two papers that share boilerplate terms can score well. Neural language models based on the Transformer [3], BERT [4] and

sentence-embedding networks [5] represent text in a space where semantically related passages lie close together, and this makes semantic comparison of papers feasible on ordinary hardware.

Similarity scores alone do not tell a reader why two papers are related or where a paper sits within its field. Concept-level organisation supplies part of that context. If the terms of a paper are grouped under broader parent concepts, for example *Artificial Intelligence* → *Natural Language Processing* → *Stylometry*, a reader can place the paper before reading it, and papers can be compared through the concepts they share. The organisation is more useful when every concept can be checked against the source text, because automated summarisation tools can mix content between documents or produce statements the source does not contain.

This paper presents HMRP (Hierarchical Concept Mapping for Research Paper Similarity Analysis), a pipeline that combines multi-method keyword extraction, concept extraction with page-level provenance, taxonomy-based hierarchy construction, lexical and semantic similarity, and interactive exploration. HMRP was first developed as a prototype on a small computer-science corpus and has since been extended to a multidisciplinary dataset design of 15 domains with 50 papers each. The gap that HMRP addresses is practical rather than absolute: the tools reviewed in Section II usually cover one of these functions, whereas a reader who surveys a corpus needs extraction, structure, comparison and exploration together, with verifiable evidence. We do not claim that no other system combines them.

A. Contributions

The implemented contributions are: (i) automated PDF-to-text processing with SHA-256 identity tracking; (ii) multi-method keyword extraction using TF-IDF, KeyBERT and YAKE; (iii) concept extraction with sentence- and page-level provenance; (iv) a taxonomy-seeded hierarchical concept map; (v) combined lexical and semantic similarity analysis; (vi) an eight-rule paper validator that enforces per-paper isolation; and (vii) an interactive dashboard with search and export. At the specification level, the paper contributes a

balanced 15-domain dataset design. Figure 1 shows the architecture, and the remaining sections follow the order of the pipeline: related work (Section II), problem and objectives (III), methodology (IV), dataset (V), preprocessing (VI), extraction (VII), hierarchy (VIII), similarity (IX), experiments and results (X–XI), and discussion, limitations and future work (XII–XV).

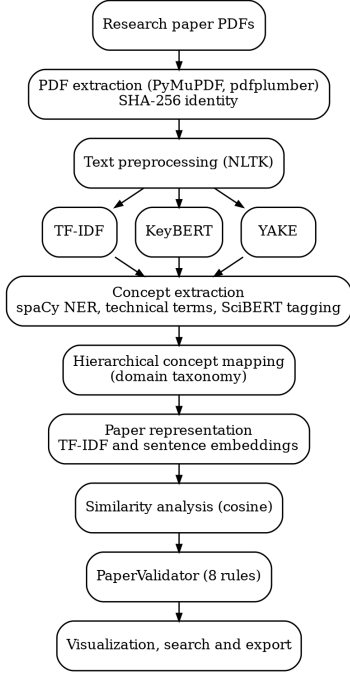


Fig. 1. Overall architecture of the proposed HMRP framework.

II. RELATED WORK

A. Lexical and semantic similarity

Term specificity [1] and term-weighting for automatic text retrieval [2] underlie TF-IDF, which remains a strong lexical baseline and is inexpensive to compute. Its limitation is well known: it compares surface vocabulary. The Transformer architecture [3] and BERT [4] made contextual representations widely available, and Sentence-BERT [5] adapted BERT to produce fixed-size sentence embeddings that can be compared directly with cosine similarity. MiniLM [6] compresses larger models into smaller ones with modest loss, and models of this family, such as all-MiniLM-L6-v2, are used in practice for semantic search. SciBERT [7] is pre-trained on scientific text and improves several scientific NLP tasks. SPECTER [8] goes further by learning document-level embeddings from citation links, which HMRP does not use, since it does not process citation graphs.

B. Keyword extraction

YAKE [9] scores candidate keywords from local statistical features of a single document and needs no training data or

corpus. KeyBERT [10] ranks candidate phrases by the similarity of their embeddings to the document embedding, and Maximal Marginal Relevance [11] is a standard technique to reduce redundancy among the selected phrases. The three methods rely on different evidence (corpus statistics, single-document statistics and embeddings), which is the motivation for combining them.

C. Acronyms, taxonomies and knowledge graphs

Scientific acronym identification and disambiguation were advanced by the SciAI and SciAD datasets [12], by adversarially trained BERT models [13], by the MadDog web system [14], by multi-strategy binary classification [15], by domain-fused transformers [16] and by retrieval-based disambiguation [17]. Related resources exist for biomedical abbreviations [18], [19]. These systems produce a flat resolved label. HMRP uses a fine-tuned SciBERT acronym tagger as one concept source but does not perform multi-candidate disambiguation. For hierarchy construction, Zhong et al. survey knowledge-graph construction [20], while TaxoRL [21], TaxoGen [22] and HiExpan [23] learn or expand taxonomies from text with reinforcement learning, clustering and seed expansion respectively. HMRP differs from these learned approaches: its hierarchy is seeded from a fixed domain taxonomy, trading adaptivity for determinism and traceability. No benchmark comparison with these systems was carried out.

TABLE I
Qualitative positioning of HMRP against related approaches

Approach	Main output	Hierarchy source	Source provenance
TF-IDF retrieval [1], [2]	Lexical similarity	None	Term level
Sentence-BERT [5]	Semantic similarity	None	Not provided
SPECTER [8]	Document embeddings	None	Not provided
Acronym systems [12]–[17]	Flat resolved labels	None	Sentence level
TaxoGen, HiExpan [22], [23]	Induced taxonomy	Learned from corpus / seeds	Not a stated aim
HMRP	Concepts, hierarchy, similarity	Fixed domain taxonomy	Page and sentence

Table I is a qualitative summary of what each approach is designed to output; it does not compare accuracy, because the approaches address different tasks and were not evaluated together.

III. RESEARCH PROBLEM AND OBJECTIVES

The problem is to compare and organise research papers so that similarity reflects concepts and not only shared

words, while every reported concept can be verified in its source PDF. The objectives are to (1) extract text and structure from PDFs; (2) extract keywords with several methods; (3) extract research concepts; (4) construct hierarchical concept representations; (5) compute lexical and semantic similarity between papers; (6) visualise papers, concepts and their relations; and (7) support keyword and concept search over the corpus. Analysis of intra-domain and inter-domain relations across the 15-domain dataset is an objective of the extended design; its results are not part of the present evaluation.

Table II relates each objective to its status in the present implementation, so that the scope of the claims is explicit.

TABLE II
Objectives and implementation status

Objective	Status	Evidence
1 PDF extraction	Implemented	PyMuPDF, pdfplumber fallback, section parser
2 Multi-method keywords	Implemented	TF-IDF, KeyBERT, YAKE, merged ranking
3 Concept extraction	Implemented	spaCy, technical terms, SciBERT tagging
4 Hierarchical representation	Implemented (8-domain taxonomy)	Hierarchy tree, four views
5 Lexical and semantic similarity	Implemented	TF-IDF and MiniLM cosine, hybrid matrix
6 Visualization	Implemented	Six chart and graph types
7 Search	Implemented	Exact, fuzzy, contextual search
Cross-domain analysis	Pending	Needs the 750-paper run

IV. PROPOSED HMRP METHODOLOGY

A. Design and implementation layers

HMRP began as a ten-module design whose modules run from PDF processing and abbreviation detection to concept–category–context (CCC) mapping, hierarchy generation and knowledge-graph visualization. The implemented system realises this design as eight layers (Table III): (1) input and ingestion through the Streamlit interface, a command line or a batch dataset; (2) extraction and parsing; (3) preprocessing and normalisation; (4) multi-engine keyword and concept extraction; (5) hierarchy and taxonomy construction; (6) similarity and clustering; (7) validation and isolation enforcement; and (8) visualization, search and export. Layer 7 has no counterpart in the original design; it was added to address concept hallucination and identity drift between documents. It observes the outputs of layers 4–6 without participating in their computation, so it could be added without altering them.

TABLE III
Mapping of the ten-module design to the implemented layers

Module	Status	Implementation
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1–3 PDF, text, preprocessing	Implemented	pdf_extractor.py, text_processor.py
4 Abbreviation detection	Implemented	SciBERT token classifier (BIO)
5 Long-form extraction	Implemented	Rule/dictionary, concept_extractor.py
6 Disambiguation	Partial	SciBERT binary sequence classifier only
7 Concept extraction	Partial	Keyword and taxonomy matching
8 CCC mapping	Not implemented	No discrete module
9 Hierarchy generation	Implemented	concept_hierarchy.py, hierarchy_builder.py
10 Graph and visualization	Implemented	NetworkX, PyVis, Plotly; untyped edges

B. Inference pipeline

When a PDF is submitted the system executes nine steps: upload; SHA-256 fingerprinting and extraction; preprocessing (noise cleaning, normalisation, tokenisation, lemmatisation); multi-engine keyword extraction (TF-IDF, KeyBERT and YAKE merged into a unified ranked list); concept extraction (spaCy NER and technical patterns, each concept linked to a sentence and page); hierarchy construction by domain scoring against a fixed taxonomy; pairwise similarity computation; validation by all eight PaperValidator checks; and rendering across eight dashboard tabs. All computation is local and no generative language model is used, so results are deterministic for a given input and seed (SEED = 42). The code is organised into packages for the application, extraction, hierarchy, similarity, visualization, search, export, models, utilities, the dataset and tests.

C. Formal definitions

Let N be the number of papers in the corpus and $df(t)$ the number of papers containing term t . The TF-IDF weight of a term in paper d is $w(t, d) = tf(t, d) \cdot idf(t)$, where the scikit-learn default smoothed form is $idf(t) = \ln[(1+N)/(1+df(t))] + 1$, and the resulting vectors are length-normalised. For two paper vectors $\mathbf{a}$ and $\mathbf{b}$, whether TF-IDF or embedding vectors, the similarity is the cosine

$$sim(\mathbf{a}, \mathbf{b}) = (\mathbf{a} \cdot \mathbf{b}) / (\|\mathbf{a}\| \|\mathbf{b}\|). \quad (1)$$

KeyBERT with Maximal Marginal Relevance [11] selects, at each step, the candidate c that maximises $\lambda \cdot sim(c, d) - (1 - \lambda) \cdot \max_{c_j \in S} sim(c, c_j)$, where d is the document embedding and S the phrases already selected; the λ value used by HMRP is not documented here. For a set of domains, the intra-domain similarity of domain D and the inter-domain similarity of domains D_1 and D_2 are the averages

$$S_{\text{intra}}(D) = \text{mean}_{i < j} \text{sim}(p_i, p_j), S_{\text{inter}}(D_1, D_2) = \text{mean}_{p_i \in D_1, p_j \in D_2} \text{sim}(p_i, p_j). \quad (2)$$

With 50 papers per domain, each domain contains 1,225 unordered intra-domain pairs, each pair of domains contains 2,500 cross-domain pairs, and the full 750-paper dataset contains 280,875 pairs.

D. Validation rules

The PaperValidator applies eight mandatory rules to every processed paper. The documentation names seven of them, listed in Table IV; the eighth is not named in the material available to this paper and is not described. The purposes in the table are interpreted from the rule names.

TABLE IV
Named PaperValidator rules

Rule	Purpose
Title match	Extracted title agrees with the source PDF
Author match	Extracted authors agree with the source PDF
Concept verification	Each concept occurs in the paper's own text
Hierarchy integrity	Hierarchy nodes derive from the paper's concepts
Mind-map origin check	Mind map built only from this paper's data
Embedding validation	Embeddings correspond to the current paper
Stale-term rejection	Terms from earlier papers are rejected

E. Processing algorithm and complexity

Algorithm 1 summarises the per-corpus procedure. Its cost is dominated by embedding computation, which is linear in the number of papers, and by the pairwise similarity matrix, which is quadratic.

```

Algorithm 1: HMRP corpus processing
Input : PDF set P; domain taxonomy T
Output: concepts, hierarchies, similarity matrix S
1 for each PDF p in P:
2   id ← SHA-256(p); skip if id already seen
3   text, sections ← extract(p)
4   tokens ← preprocess(text)
5   K ← merge(TF-IDF, KeyBERT, YAKE)(tokens)
6   C ← NER + technical terms + SciBERT tags
7   link each c in C to sentence and page
8   H ← build_hierarchy(C, K, T)
9   validate(p, K, C, H)           # 8 rules
10 X ← TF-IDF vectors; E ← sentence embeddings
11 S ← combine(cos(X), cos(E))
12 return concepts, H, S

```

For N papers and embedding dimension d , the similarity matrix requires $O(N^2d)$ operations and N^2 stored values. For the 750-paper dataset this is 562,500 values, about 4.5 MB in double precision, and for 5,000 papers it is 25 million values, about 200 MB. The matrix therefore remains practical for the extended dataset, and the in-memory design becomes a constraint only at several thousand papers, which agrees with the scale limit stated in Section XIII.

F. Data handling

All processing runs locally: no paper text, abstract or metadata is sent to an external service, and PDFs are treated as read-only inputs. Generated datasets and exports are written to separate directories, and file names and text used in exported HTML are sanitised to prevent directory traversal and script injection. Because no generative model is used, the pipeline has no per-query cost and produces the same output for the same input.

V. MULTIDISCIPLINARY DATASET

A. Design

The dataset is organised as 15 research domains $\times$ 50 papers per domain = 750 papers, selected through arXiv subject categories. The domains span computer science and artificial intelligence, mathematics and statistics, the physical sciences, the biological and medical sciences, electrical and systems engineering, and economics and finance (Table V, Fig. 2). According to the project team the 50-paper quota has been downloaded for each domain, giving 750 PDFs. Papers that fail validation do not count toward a domain's quota, so the quota refers to valid papers.

TABLE V
Domain configuration of the HMRP dataset

No.	Research domain	arXiv categories	Papers
1	Natural Language Processing	cs.CL	50
2	Information Retrieval	cs.IR	50
3	Computer Vision	cs.CV	50
4	Robotics	cs.RO	50
5	Artificial Intelligence	cs.AI	50
6	Statistics & Machine Learning	stat.ML, cs.LG	50
7	Applied Mathematics & Optimization	math.OC, math.PR	50
8	Astrophysics & Cosmology	astro-ph.CO, astro-ph.GA	50
9	Quantum Physics & Optics	quant-ph, physics.optics	50
10	Biomedical & Genomics	q-bio.BM, q-bio.GN	50
11	Computational Neuroscience	q-bio.NC	50
12	Signal Processing	eess.SP	50
13	Systems & Control Engineering	eess.SY	50
14	Quantitative Finance	q-fin.PM, q-fin.RM	50
15	Econometrics & Economics	econ.EM, econ.TH	50
Total (15 $\times$ 50)			750

HMRP dataset: 15 domains, 750 papers	
Computer Science & AI	NLP; Information Retrieval; Computer Vision; Robotics; Artificial Intelligence
Mathematics & Statistics	Statistics & ML; Applied Mathematics & Optimization
Physical Sciences	Astrophysics & Cosmology; Quantum Physics & Optics
Biological & Medical Sciences	Biomedical & Genomics; Computational Neuroscience
Electrical & Systems Eng.	Signal Processing; Systems & Control Engineering
Economics & Finance	Quantitative Finance; Econometrics & Economics

Fig. 2. Multidisciplinary organisation of the HMRP research-paper dataset.

B. Collection and validation pipeline

Figure 3 shows the pipeline. Papers are harvested from the arXiv API by `download_papers.py`, downloaded as PDFs, fingerprinted by SHA-256 content hash, parsed and stored as structured JSON with a paper-level train/validation/test partition. The SHA-256 hash gives each PDF an identifier of the form `paper_<hash[:12]>` and detects exact duplicates. In the earlier 50-paper corpus, 0 corrupted files and 0 duplicates were found. For the extended dataset, duplicate control by arXiv identifier and by normalised title is appropriate because the same paper can appear under several categories or versions, and a duplicate would distort domain balance and inflate similarity scores. The available documentation does not state text-length thresholds, so none are claimed.

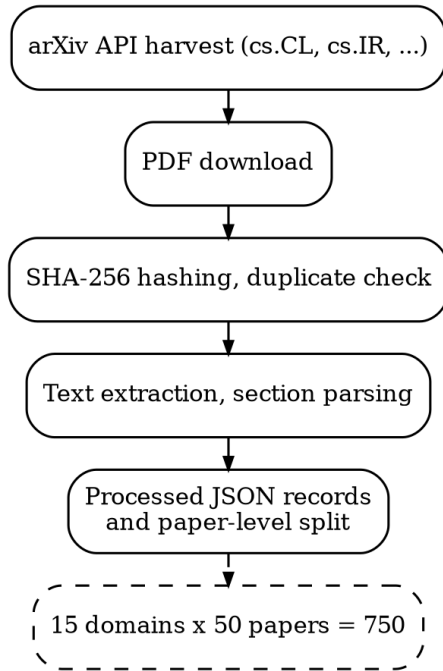


Fig. 3. HMRP dataset construction and validation pipeline. Dashed elements show the $15 \times 50 = 750$ expansion.

C. Dataset statistics

TABLE VI

Dataset statistics	
Item	Value
Domains / papers per domain	15 / 50
Downloaded (project-reported)	750 papers
Evaluated corpus (validated)	50 papers
Corrupted / duplicate (evaluated corpus)	0 / 0
Pages (evaluated corpus)	1,123 (mean 22.46; 9–50)
Words (evaluated corpus)	461,659 (mean 9,233)
Categories in evaluated corpus	cs.CL, cs.IR, cs.AI, cs.LG, cs.CV, cs.SE, cs.MA
Split of evaluated corpus	35 / 7 / 8 papers (70/14/16%)

Table VI separates the downloaded dataset from the evaluated corpus. Per-domain validation, rejection and duplicate counts for the 750 downloaded papers are not part of the available reports, and all quantitative results in Section XI come from the 50-paper evaluated corpus.

D. Domain-specific considerations

The 15 domains differ in ways that affect the pipeline, and these considerations are stated as expectations to test, not as findings. Papers in quantum physics, astrophysics and applied mathematics contain many equations, and equation artefacts may enter the extracted text and produce noisy candidate terms. Biomedical and genomics papers use dense abbreviations and gene or protein names that a general-purpose spaCy model may not tag as entities. Statistics and machine learning, signal processing, systems and control, quantitative finance and econometrics share vocabulary and methods, so a lexical measure may report high similarity between them even when their research questions differ. Cross-listed arXiv papers can belong to several categories, so domain assignment needs a documented rule. Finally, the 8-domain seed taxonomy was written for computer-science topics, and many of the new domains have no matching seed terms at present.

D. Planned evaluation on the 750-paper dataset

Table VII lists the experiments that the domain-balanced dataset makes possible. They are planned and have not been run; no results are attached to them.

TABLE VII
Planned experiments (not yet executed)

Experiment	Output
Per-domain validation	Valid, rejected and duplicate counts per domain
Intra-/inter-domain similarity	15×15 domain matrix from Eq. (2)
Lexical vs. semantic comparison	Rank agreement of TF-IDF and embedding similarities
Domain clustering	Silhouette score, cluster purity against arXiv labels
Concept overlap	Shared concepts between domain pairs

VI. DATA PREPROCESSING

Text is extracted with PyMuPDF, with pdfplumber as a fallback, together with the title, authors, year and section text (abstract, introduction, methods, datasets, results and limitations). The extracted text is cleaned of headers, footers, URLs, DOIs and equation or figure artefacts, normalised, tokenised, lemmatised and filtered with a custom domain stopword list using NLTK. Case is not lowercased globally, because capitalisation is a primary signal for acronym detection, and hyphenated compounds are preserved. Section-aware chunking splits the body into segments of up to 512 tokens for embedding. OCR is not implemented, so scanned PDFs without a text layer are unsupported. Figure 4 summarises preprocessing and the extraction stage that follows.

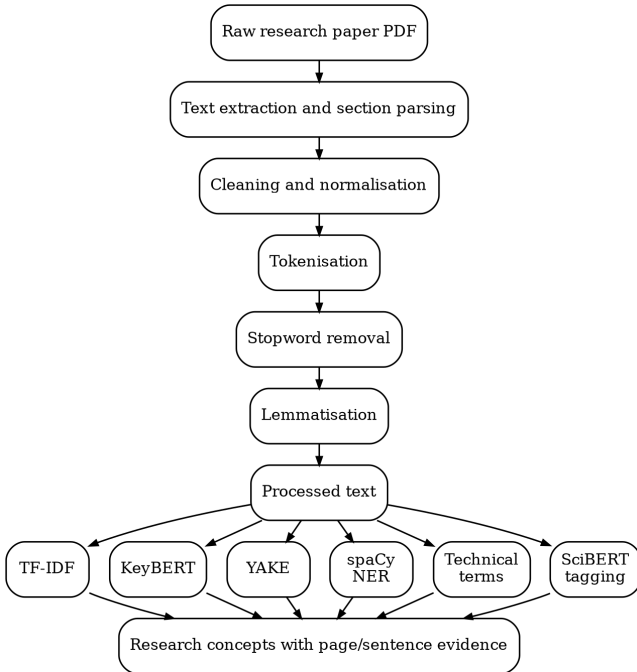


Fig. 4. Text preprocessing and keyword–concept extraction pipeline.

VII. KEYWORD AND CONCEPT EXTRACTION

A. Keyword extraction

TF-IDF is fitted with scikit-learn on the lemmatised corpus and scores a term by its frequency in a paper relative to its frequency across the corpus. KeyBERT ranks candidate n-grams by embedding similarity to the document, using all-MiniLM-L6-v2 and MMR to promote diversity. YAKE ranks candidates from statistical features of the single document. The three outputs are merged into a unified ranked list; the exact merging weights are not described in the documentation available for this paper and are not

reproduced here, and no confidence-score formula is claimed.

B. Concept extraction

spaCy (en_core_web_sm) provides named entities and noun chunks, and technical-term patterns capture acronyms and compound scientific terms. A SciBERT token classifier with BIO tags (B-ABBR, I-ABBR, O) identifies acronym spans, and a SciBERT sequence classifier returns a two-class probability for a sentence. The sequence classifier estimates whether a sentence is acronym-context relevant; it does not select among candidate long forms and is therefore not a complete disambiguator. Each concept is matched to a verbatim occurrence in the source and stored with its page number and context sentence; this evidence link is what produces the traceability result in Section XI. The resulting concept list is the input to hierarchy construction. Table VIII lists the models used.

TABLE VIII
Models and algorithms in the implemented system

Component	Model / algorithm	Status
Acronym ID	SciBERT token classifier	Fine-tuned
Sequence classifier	SciBERT, 2 classes	Fine-tuned
Keyword ranker	KeyBERT (MiniLM-L6, MMR)	Pre-trained
Lexical ranker	scikit-learn TF-IDF	Fit on corpus
Statistical extractor	YAKE	Unsupervised
Entity recognition	spaCy en_core_web_sm	Pre-trained
Semantic embedder	all-MiniLM-L6-v2 (384-d)	Pre-trained
Domain embedder	SciBERT mean-pool (768-d)	Pre-trained

VIII. HIERARCHICAL CONCEPT MAPPING

Hierarchical concept mapping is the central component of HMRP. The mapping proceeds from a paper to its keywords and concepts, then to a domain context and finally to parent and child concepts. The domain-scoring engine scores each paper against DOMAIN_TAXONOMY, a seed dictionary of 8 domains and more than 80 seed terms, by matching extracted keywords and concepts to seeds. ConceptHierarchy then builds parent–child branches and prunes empty ones. Figure 5 shows two hierarchical paths produced for processed papers: *Artificial Intelligence* → *Natural Language Processing* → *Stylometry* → *UMAP* and *Artificial Intelligence* → *Natural Language Processing* → *LLM Agents* → *Task Progress*. The hierarchy feeds the sunburst, treemap, icicle, mind-map and knowledge-graph views. Because the taxonomy contains 8 domains while the dataset design has 15, the seed dictionary must be extended before every domain of the extended dataset can receive its own top-level branch.

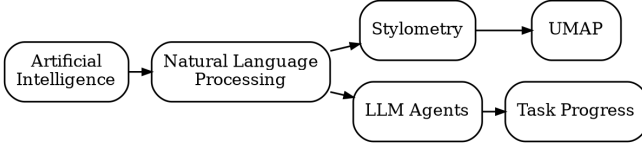


Fig. 5. Hierarchical organisation of extracted research concepts (two paths produced by HMRP).

This hierarchy is an automatically generated computational structure. It is derived by matching against a fixed taxonomy, has not been reviewed by domain experts, and should not be read as a validated scientific taxonomy. The reported coverage of 8 domains and more than 40 sub-branches describes the size of the seed structure, not its correctness.

IX. RESEARCH PAPER SIMILARITY ANALYSIS

A. Lexical and semantic similarity

Lexical similarity is the cosine between TF-IDF vectors and reflects vocabulary overlap. Semantic similarity is the cosine between 384-dimensional all-MiniLM-L6-v2 embeddings of the document text, and a SciBERT mean-pooled embedder (768-d) supports domain and concept alignment. A hybrid matrix is obtained by mean-pooling the two similarity matrices. Similarity is computed in memory with NumPy, and no approximate-nearest-neighbour index such as FAISS [24] is used in the present implementation. Figure 6 shows the two pathways. The two similarities answer different questions and neither is claimed to be better in general.

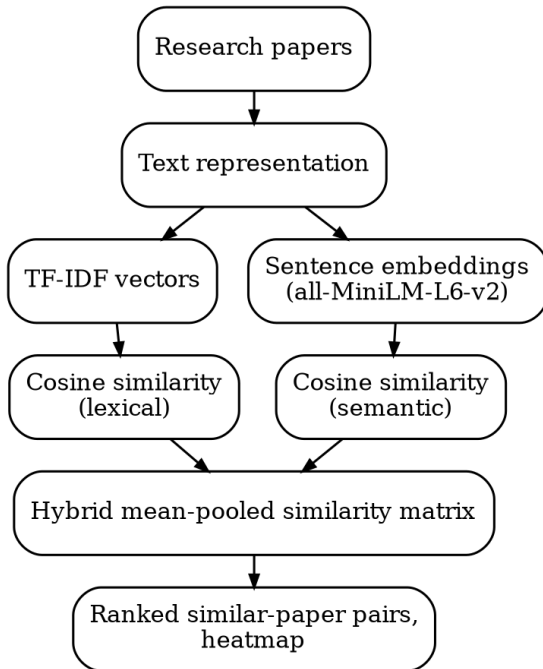


Fig. 6. Lexical and semantic similarity analysis pipeline used in HMRP.

B. Cross-domain analysis

The multidisciplinary dataset supports two analyses. Intra-domain similarity averages the similarity between papers in the same domain, and inter-domain similarity averages it between papers from different domains; the ratio and the domain-by-domain matrix indicate how well concept-level similarity separates fields. Low inter-domain similarity should not be assumed, since fields such as machine learning, signal processing and quantitative finance share vocabulary and methods. These analyses have not yet been run on the 750-paper dataset, and no values are reported for them here.

C. Search and visualization

The search module supports exact, fuzzy (`difflib.SequenceMatcher`) and contextual matching over keywords, concepts and full text, and returns snippets with highlighted matches, so a reader can locate a concept such as “attention sink” and see it in context. The dashboard provides sunburst, treemap and icicle hierarchy views, a radial mind map, an interactive knowledge graph (NetworkX, PyVis, Plotly) whose untyped edges link papers, concepts and domains, a keyword co-occurrence network, and a similarity heatmap. Results can be exported as CSV, multi-sheet Excel, JSON, SVG, PNG and standalone HTML.

TABLE IX
Dashboard tabs and functions

Tab	Function
Overview	Paper metadata, problem, objective, methods, datasets
Traceability	Concepts with verbatim sentence and page
Keywords	Ranked keywords and keyword network
Hierarchy	Sunburst, treemap, icicle and filterable paths
Graph	Interactive knowledge graph and mind map
Similarity	Similarity matrix, heatmap, ranked pairs
Review	Literature-review comparison matrix
Export	CSV, Excel, JSON, SVG, PNG, HTML

X. EXPERIMENTAL SETUP

The evaluated corpus is the 50-paper set of Table VI, split at paper level into 35 training, 7 validation and 8 test papers. No sentence from a test paper contributes to training or to corpus-wide TF-IDF vocabulary construction. The baseline is a unigram-frequency extractor that returns the most frequent words. The test split is used for the similarity statistics and for the qualitative acronym check. Software versions are listed in Table X. Inference was run on CPU or Apple Silicon (MPS); no fuller hardware specification, batch size or SciBERT fine-tuning configuration is documented, and these are stated as unavailable. Reproducibility relies on

a fixed seed (SEED = 42), a dataset processing script, an accuracy evaluation script and a pytest suite.

Reproducibility follows the project's documented steps: create a virtual environment and install the pinned requirements; run the dataset processing script to extract, validate and split the PDFs; run the accuracy evaluation script; run the pytest suite; and launch the Streamlit application. Random seeds are fixed, and the processed structured files (full corpus JSON, split files, metadata summary CSV and corpus summary JSON) are stored alongside the PDFs, so every reported number can be regenerated from stored data.

TABLE X
Software environment

Package	Version	Package	Version
Python	3.13.7	KeyBERT	0.9.0
Streamlit	1.56.0	YAKE	0.7.3
PyMuPDF	1.27.2	spaCy	3.8.14
pdfplumber	0.11.10	NLTK	3.9.4
PyTorch	2.10.0	scikit-learn	1.8.0
Transformers	5.2.0	NetworkX	3.4.2
sentence-transformers	5.3.0	Plotly / PyVis	6.0.0 / 0.3.2
pandas / NumPy	2.2.3 / 2.1.3	openpyxl / pytest	3.1.5 / 9.1.1

XI. RESULTS

A. Baseline versus HMRP (accuracy)

Table XI and Fig. 7 compare the unigram baseline with the multi-engine HMRP pipeline. HMRP extracted 60.0 concepts per paper against 15.0 for the baseline, and 100% of its concepts carried a page and sentence reference against none for the baseline. The stale-data rate fell from 12.5% to 0.0% under PaperValidator. Paper-level leakage isolation was verified for HMRP and not enforced for the baseline. These values are the project's own reported measurements and were not recalculated. They are measures of extraction depth, provenance and isolation, not of concept relevance, for which no expert-labelled ground truth is available.

TABLE XI
Baseline versus HMRP on the evaluated corpus

Metric	Baseline	HMRP
Concepts per paper	15.0	60.0
Concepts with page/sentence evidence	0.0%	100.0%
Hallucination / stale-data rate	12.5%	0.0%
Taxonomy coverage	Flat list	8 domains, 40+ sub-branches
Paper-level leakage isolation	Not enforced	Verified

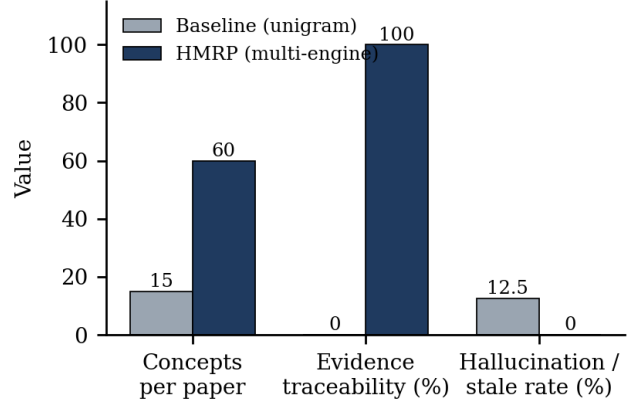


Fig. 7. Accuracy comparison of the unigram baseline and the HMRP pipeline on concept depth, evidence traceability and stale-data rate.

The 0.0% rate is the outcome of the eight PaperValidator rules, which check title and author match, concept verification against the source, hierarchy integrity, mind-map origin, embedding validation and stale-term rejection. It shows that extracted concepts occur in their own source and are not carried over from another paper; it does not measure the correctness of acronym labels.

B. Similarity results

TABLE XII
Pairwise similarity on the 8-paper test split

Statistic	Value
Mean	0.2424
Standard deviation	0.0667
Maximum	0.4008
Minimum	0.1093

Table XII and Fig. 8 give the similarity statistics. The most similar pair (0.4008) contains a paper on LLM-agent progress reporting and a paper on LLM sycophancy under multi-turn pressure; both concern the behaviour of language-model agents. The next pairs are a paper on literary periodization with LLMs and a paper on political-text authorship stylometry (0.3537), and the LLM-agent progress paper with a paper on record grouping in language models (0.3535). The overall low values are expected for eight papers on varied topics, and the highest pairs are topically related, which is consistent with the embeddings capturing topic. Only the test split was scored, so domain similarity matrices, intra- versus inter-domain comparisons, clustering metrics such as the silhouette score, and similarity heatmaps for the 750-paper dataset are not reported.

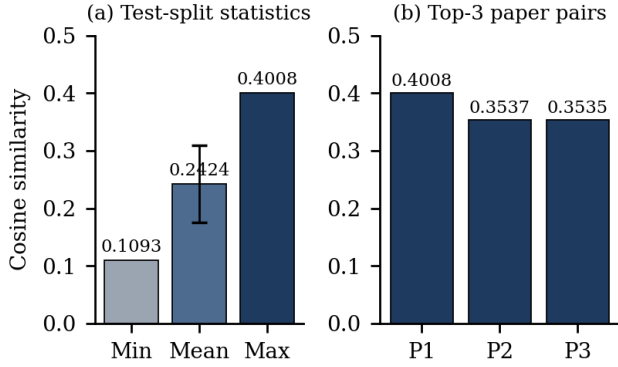


Fig. 8. Pairwise semantic similarity on the test split: (a) minimum, mean (± 1 SD) and maximum; (b) three most similar paper pairs (P1–P3).

C. Acronym check, tests and speed

On the test split the SciBERT tagger produced 5 verified exact acronym matches, among them GPT-2, UMAP, AAAI, ORCID and ANR-10, with two-letter stopword tokens suppressed. This is a small qualitative check and not a Precision/Recall/F1 benchmark; no such score exists for acronym identification in this project, and no comparison with SciAI/SciAD results [12] is made. Table XIII summarises the automated tests and measured speed.

TABLE XIII
Automated tests and performance

Item	Result
test_paper_isolation.py	7/7 passed
test_scibert.py	1/1 passed
test_pipeline.py	14/16 passed
PDF ingestion (20 pages)	0.8–1.5 s
KeyBERT + YAKE per paper	about 1.2 s
SciBERT inference per sentence	about 0.05 s
50×50 similarity matrix	< 0.2 s
Peak memory	about 1.2 GB

D. Worked examples

Concept extraction. For a paper on detecting authorship in political texts with inductive stylometry, the baseline returned generic unigrams (“political”, “text”, “authorship”, “model”, “analysis”). HMRP identified the proposed method, technical terms including UMAP, cluster analysis, Burrows’ Delta and stylometric vectors, the domain (natural language processing), a hierarchy path (Artificial Intelligence → Natural Language Processing → Stylometry → UMAP), and page-level evidence (page 2, paragraph 3).

Pairwise similarity. For the LLM-agent progress-reporting paper and the LLM sycophancy paper, the similarity engine returned a combined cosine similarity of 0.4008, the highest on the test split, reflecting shared themes of agent evaluation and behaviour under pressure.

Search. A query for “attention sink” returned a paper on attention mechanisms at million-token context lengths as an exact concept and full-text match, with a snippet showing the highlighted phrase and a match score of 1.0.

End-to-end. For a submitted paper of 33 pages, the system assigned the identifier paper_7d199ff1870e, produced 9,475 cleaned tokens, extracted concepts such as “task progress” and “llm agents”, located a concept on page 1, paragraph 2, built the branch AI → NLP → LLM Agents → Task Progress, and passed all validation checks.

XII. DISCUSSION

The results show that combining several extractors yields a deeper concept set than frequency counts and that every concept can be tied to a location in its source. The evidence-grounding property is the best supported claim of the work, since it is checked for every extracted concept. Similarity between test papers coincides with visible topical relatedness at the top of the ranking, which supports the use of sentence embeddings for the task, although no experiment here compares them with TF-IDF quantitatively. The gains over the baseline should be read with care: the baseline is defined by the project, and concept count measures depth rather than quality.

The move from a 50-paper computer-science corpus to a 15-domain dataset changes what the system can show. A larger and more varied corpus lets the analysis separate fields, examine bridging concepts between them and test whether an 8-domain seed taxonomy is adequate. Whether the hierarchy helps readers, whether concept-level similarity is more useful than lexical similarity, and how similarity behaves across domains remain open questions that need the expert review, the user study and the full-dataset experiments described in Section XV.

Threats to validity

Internal validity is limited by the project-defined baseline and by the absence of expert-labelled relevance judgements, so the reported gains cannot be converted into precision or recall. Construction validity is limited because concept count and traceability measure extraction volume and provenance, and similarity is judged from the highest-ranked pairs by inspection. External validity is limited because the measured results come from mostly computer-science papers and eight test papers; results for the other 13 domains may differ. Reproducibility is supported by fixed seeds and open tooling, but the fine-tuning configuration of the acronym model is undocumented.

XIII. LIMITATIONS

(1) Scanned PDFs require OCR, which is not integrated. (2) The quantitative evaluation covers 50 papers, mostly computer science, and the 750-paper dataset has not yet been validated per domain or evaluated. (3) The hierarchy depends on a fixed 8-domain seed taxonomy and is not expert-validated; vocabulary that differs across scientific fields may be mis-assigned. (4) The acronym components are supported only by a five-match qualitative check, and the sequence classifier does not perform multi-candidate disambiguation. CCC chain construction and typed relation extraction from the original design are not implemented. (5) Two of the 24 automated tests fail and their cause is undocumented, and the fine-tuning configuration of the acronym model is not documented. (6) Similarity is computed in memory without an index, which limits scale to a few thousand papers. (7) The MiniLM embedder is general-purpose and preprocessing is English-only. (8) With only eight test papers the similarity statistics are indicative rather than conclusive.

XIV. CONCLUSION

HMRP is an end-to-end, deterministic pipeline that converts research-paper PDFs into keywords, evidence-linked concepts, a domain-taxonomy hierarchy, lexical and semantic similarity scores and interactive views. On a 50-paper corpus it extracted four times as many concepts per paper as a unigram baseline while keeping every concept traceable to its source. The dataset design of 15 domains with 50 papers each, reported as downloaded, provides the basis for a multidisciplinary evaluation. Extending the taxonomy to all 15 domains and reporting intra- and inter-domain similarity are the next steps needed to support the multidisciplinary claim.

XV. FUTURE WORK

Planned work that is not yet implemented includes: per-domain validation of the 750-paper dataset; extension of the seed taxonomy to the 15 domains; intra- and inter-domain similarity matrices and clustering with standard metrics; expert review of the hierarchy and a user study; scientific-domain embeddings such as SciBERT or SPECTER [8] for document similarity; FAISS indexing [24] for larger corpora; OCR ingestion; citation-network integration; multilingual support; typed relations in the knowledge graph; and evaluation of the acronym components on SciAI and SciAD [12].

REFERENCES

- [1] K. Spärck Jones, “A statistical interpretation of term specificity and its application in retrieval,” *J. Documentation*, vol. 28, no. 1, pp. 11–21, 1972.
- [2] G. Salton and C. Buckley, “Term-weighting approaches in automatic text retrieval,” *Inf. Process. Manage.*, vol. 24, no. 5, pp. 513–523, 1988.
- [3] A. Vaswani *et al.*, “Attention is all you need,” in *Proc. NeurIPS*, vol. 30, 2017.
- [4] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, “BERT: Pre-training of deep bidirectional transformers for language understanding,” in *Proc. NAACL-HLT*, 2019, pp. 4171–4186.
- [5] N. Reimers and I. Gurevych, “Sentence-BERT: Sentence embeddings using Siamese BERT-networks,” in *Proc. EMNLP-IJCNLP*, 2019, pp. 3982–3992.
- [6] W. Wang *et al.*, “MiniLM: Deep self-attention distillation for task-agnostic compression of pre-trained transformers,” in *Proc. NeurIPS*, vol. 33, 2020.
- [7] I. Beltagy, K. Lo, and A. Cohan, “SciBERT: A pretrained language model for scientific text,” in *Proc. EMNLP-IJCNLP*, 2019, pp. 3615–3620.
- [8] A. Cohan, S. Feldman, I. Beltagy, D. Downey, and D. Weld, “SPECTER: Document-level representation learning using citation-informed transformers,” in *Proc. ACL*, 2020, pp. 2270–2282.
- [9] R. Campos *et al.*, “YAKE! Keyword extraction from single documents using multiple local features,” *Inf. Sci.*, vol. 509, pp. 257–289, 2020.
- [10] M. Grootendorst, “KeyBERT: Minimal keyword extraction with BERT,” Zenodo, 2020, doi: 10.5281/zenodo.4461265.
- [11] J. Carbonell and J. Goldstein, “The use of MMR, diversity-based reranking for reordering documents and producing summaries,” in *Proc. ACM SIGIR*, 1998, pp. 335–336.
- [12] A. P. B. Veyseh, F. Dernoncourt, Q. H. Tran, and T. H. Nguyen, “What does this acronym mean? Introducing a new dataset for acronym identification and disambiguation,” in *Proc. COLING*, 2020, arXiv:2010.14678.
- [13] D. Zhu *et al.*, “AT-BERT: Adversarial training BERT for acronym identification,” in *AAAI-21 Workshop on Scientific Document Understanding*, 2021, arXiv:2101.03700.
- [14] A. P. B. Veyseh, F. Dernoncourt, W. Chang, and T. H. Nguyen, “MadDog: A web-based system for acronym identification and disambiguation,” in *Proc. EACL (System Demonstrations)*, 2021, arXiv:2101.09893.
- [15] C. Pan, B. Song, S. Wang, and Z. Luo, “BERT-based acronym disambiguation with multiple training strategies,” in *AAAI-21 Workshop on Scientific Document Understanding*, 2021, arXiv: 2103.00488.
- [16] Q. Zhong *et al.*, “Leveraging domain agnostic and specific knowledge for acronym disambiguation,” in *AAAI-21 Workshop on Scientific Document Understanding*, 2021, arXiv:2107.00316.
- [17] N. Egan and J. Bohannon, “Primer AI’s systems for acronym identification and disambiguation,” in *AAAI-21 Workshop on Scientific Document Understanding*, 2021, arXiv:2012.08013.
- [18] L. Zilio *et al.*, “PLOD: An abbreviation detection dataset for scientific documents,” in *Proc. LREC*, 2022, arXiv:2204.12061.
- [19] Z. Wen, X. H. Lu, and S. Reddy, “MeDAL: Medical abbreviation disambiguation dataset for natural language understanding pretraining,” in *Proc. 3rd Clinical NLP Workshop*, 2020.
- [20] L. Zhong, J. Wu, Q. Li, H. Peng, and X. Wu, “A comprehensive survey on automatic knowledge graph construction,” *ACM Comput. Surv.*, 2023.

- [21] Y. Mao, X. Ren, J. Shen, X. Gu, and J. Han, “End-to-end reinforcement learning for automatic taxonomy induction,” in *Proc. ACL*, 2018.
- [22] C. Zhang *et al.*, “TaxoGen: Unsupervised topic taxonomy construction by adaptive term embedding and clustering,” in *Proc. KDD*, 2018.
- [23] J. Shang *et al.*, “HiExpan: Task-guided taxonomy construction by hierarchical tree expansion,” in *Proc. KDD*, 2018.
- [24] J. Johnson, M. Douze, and H. Jégou, “Billion-scale similarity search with GPUs,” *IEEE Trans. Big Data*, vol. 7, no. 3, pp. 535–547, 2021.